Modeling Neutron Stars
Lecture-1

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LPC, Université Caen Normandie
https://sites.google.com/view/nucleartheory/home

UNIVERSITÉ CAEN NORMANDIE

Nuclear Theory Group
December 19, 2020

What to expect from the course

"Blessed is he who expects nothing, for he shall never be disappointed."-Alexander Pope

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- Equation of State

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- Introduction to General Theory of Relativity
- Tolman-Oppenheimer-Volkoff Equation
- Equation of State
- Probing the Structure of Neutron Star

What we will cover today

Introduction to Neutron Stars

- Life Cycle of a Star
- Death of a Star
- Why field theory?
- Necessity of General Theory of Relativity
- Tolman Oppenheimer Volkoff (TOV) equation
- Unification of Equation of state

Timeline of universe

All started with a Big Bang !!

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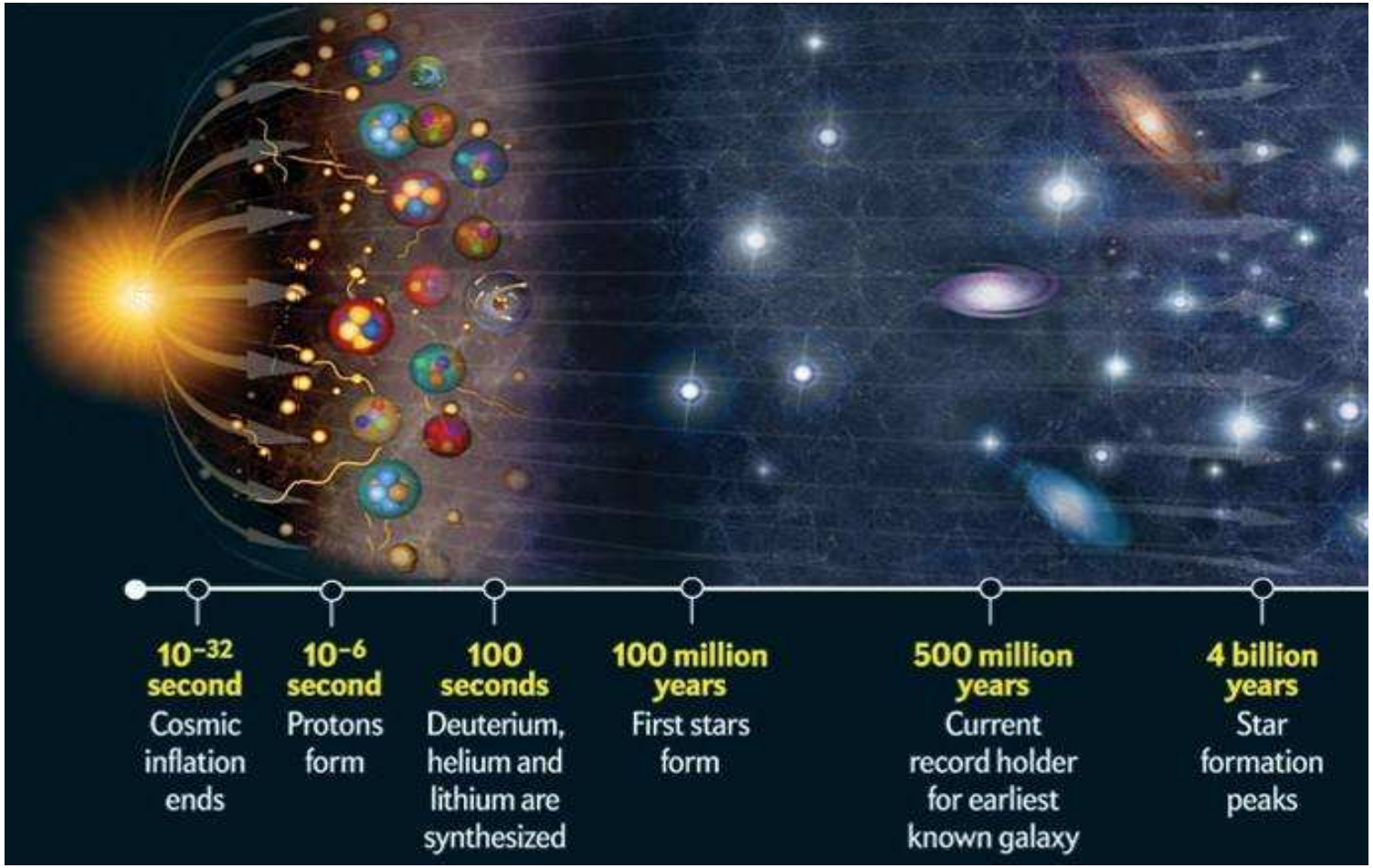

https://knowthyself.forumotion.net/t2464-hellenic-pantheon

Constituents of universe

Dark and visible matter

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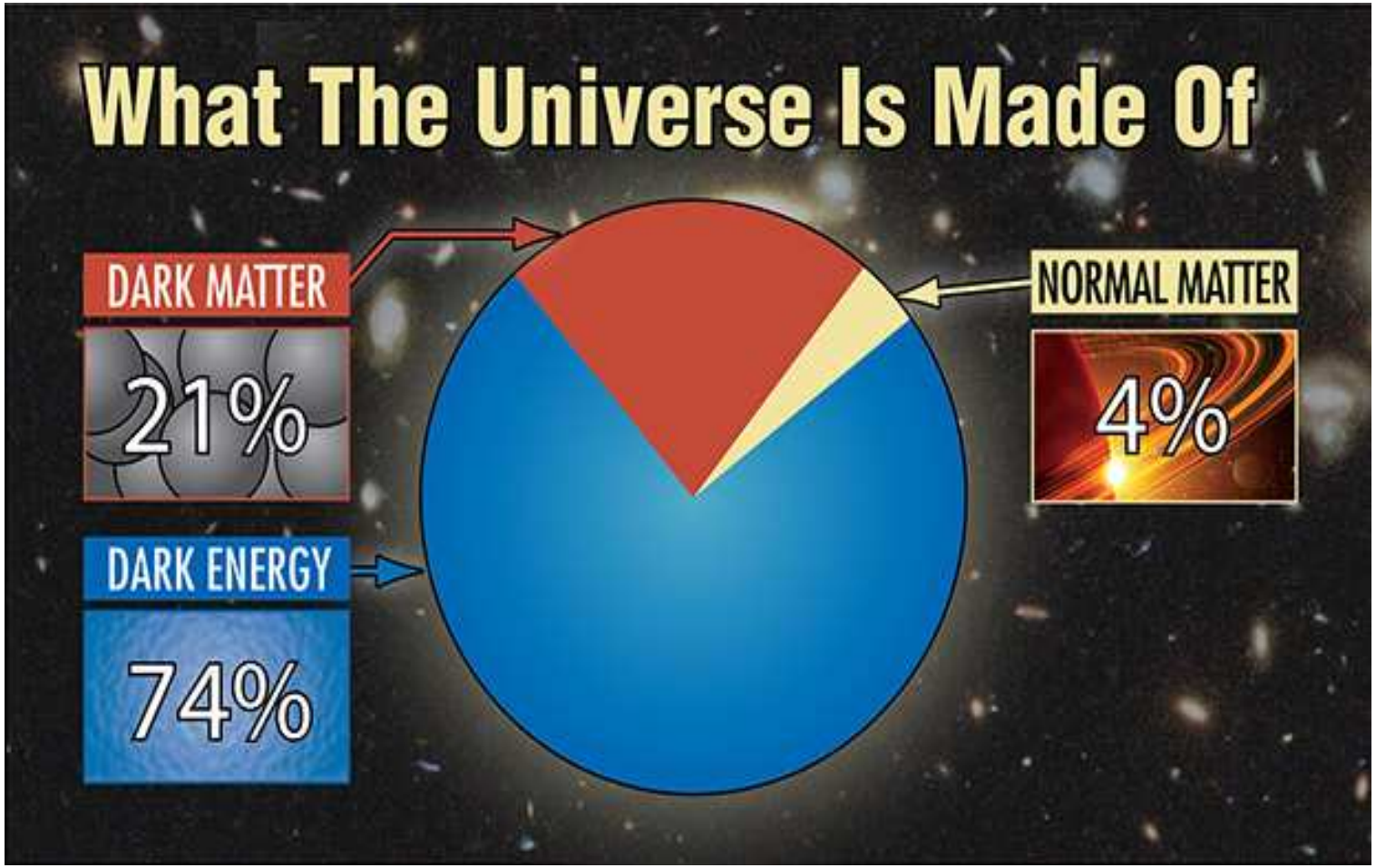
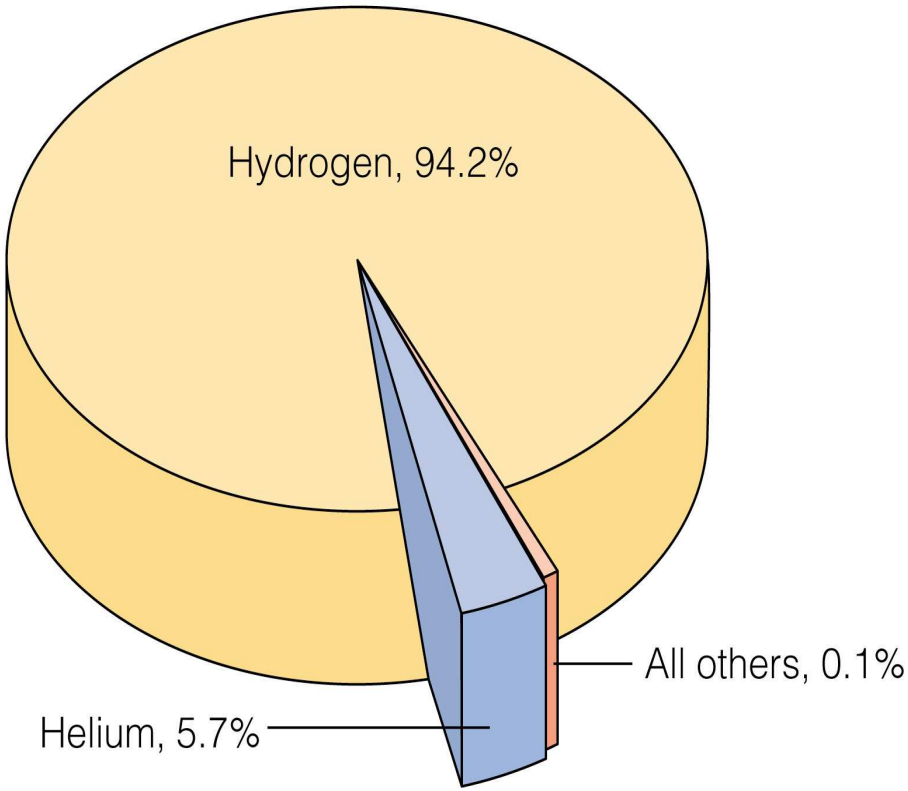


https://hetdex.org/dark_energy/dark_matter.html

Observable Universe

Abundance Curve



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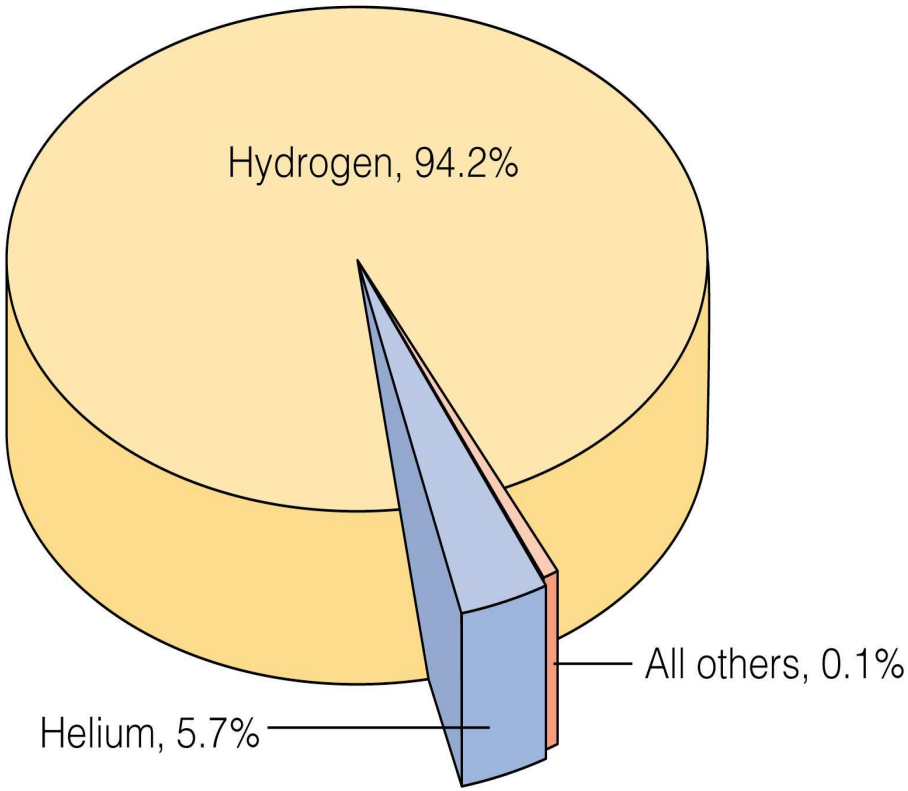

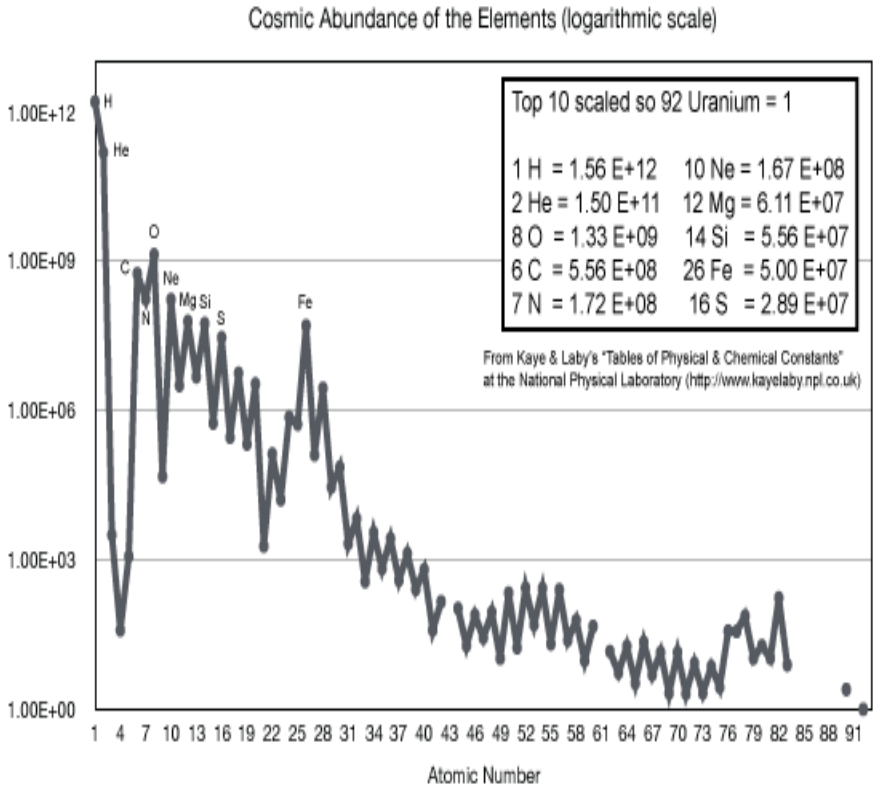

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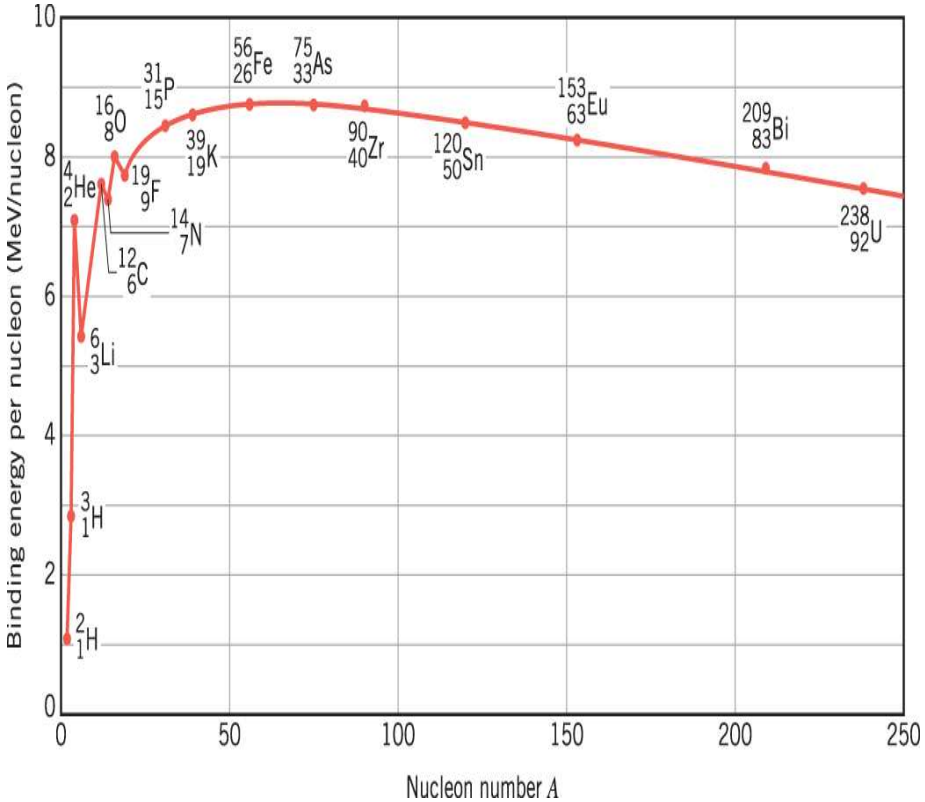
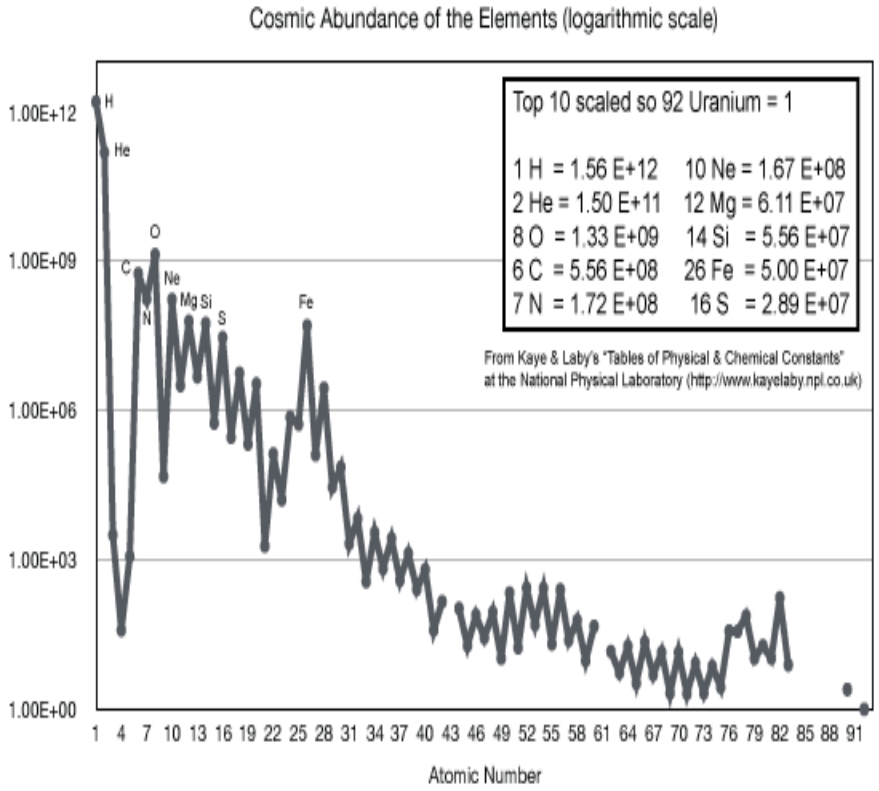
Observable Universe

Binding Energy Curve





https://www.quora.com/Why-does-nuclear-fusion-release-more-energy-than-nuclear-fission

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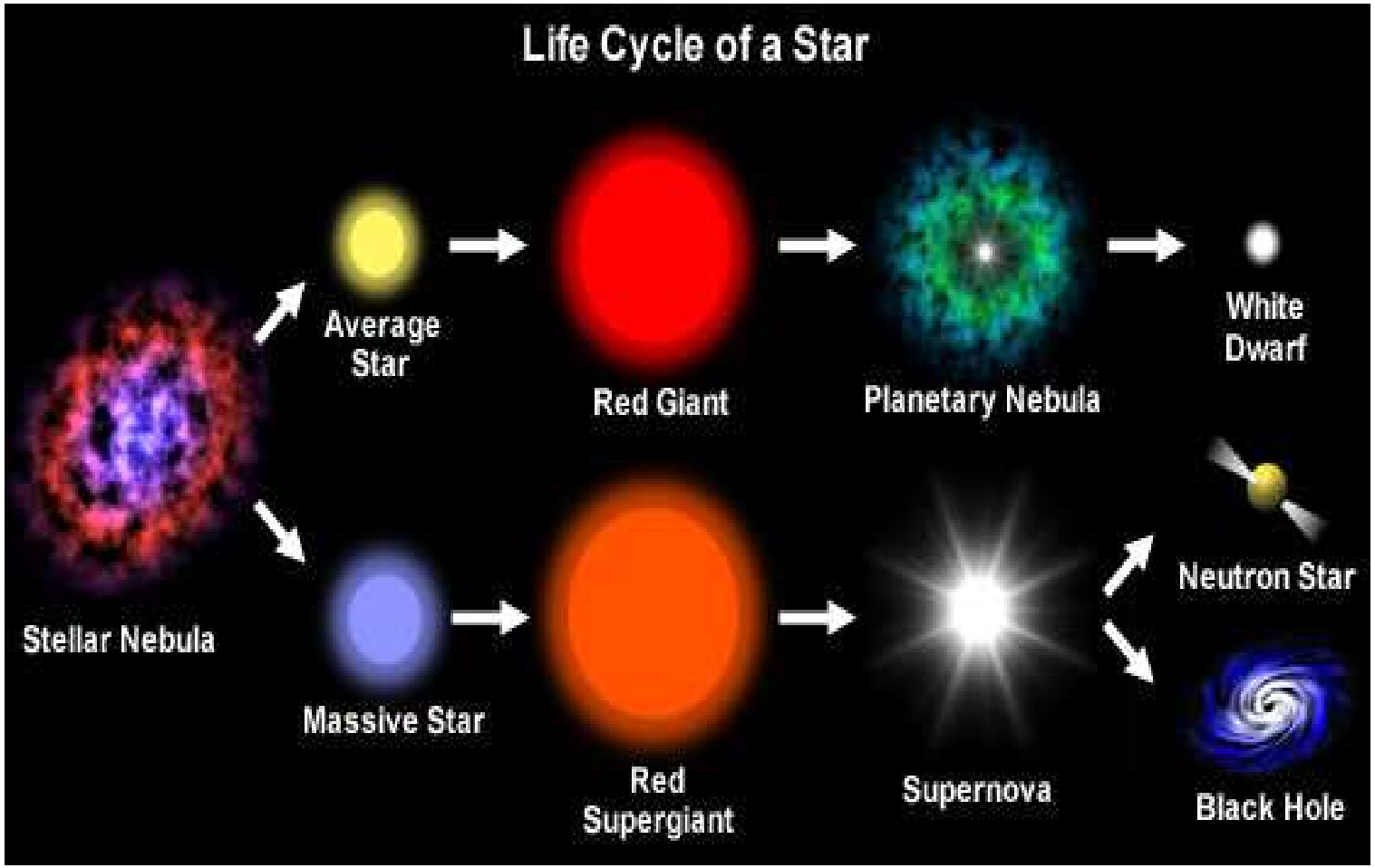
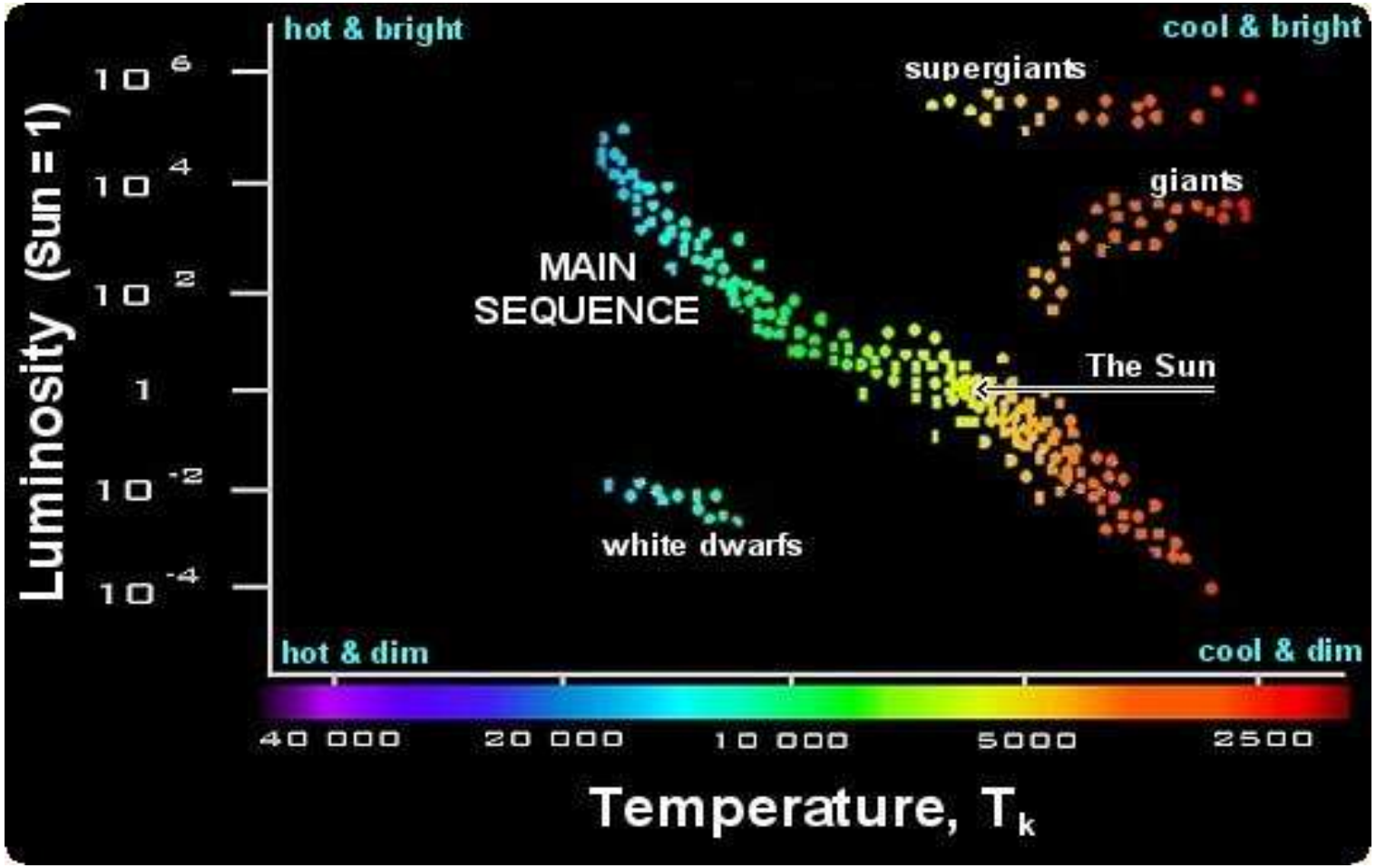
Life Cycle of a Star



https://www.schoolsobservatory.org/learn/astro/stars/cycle

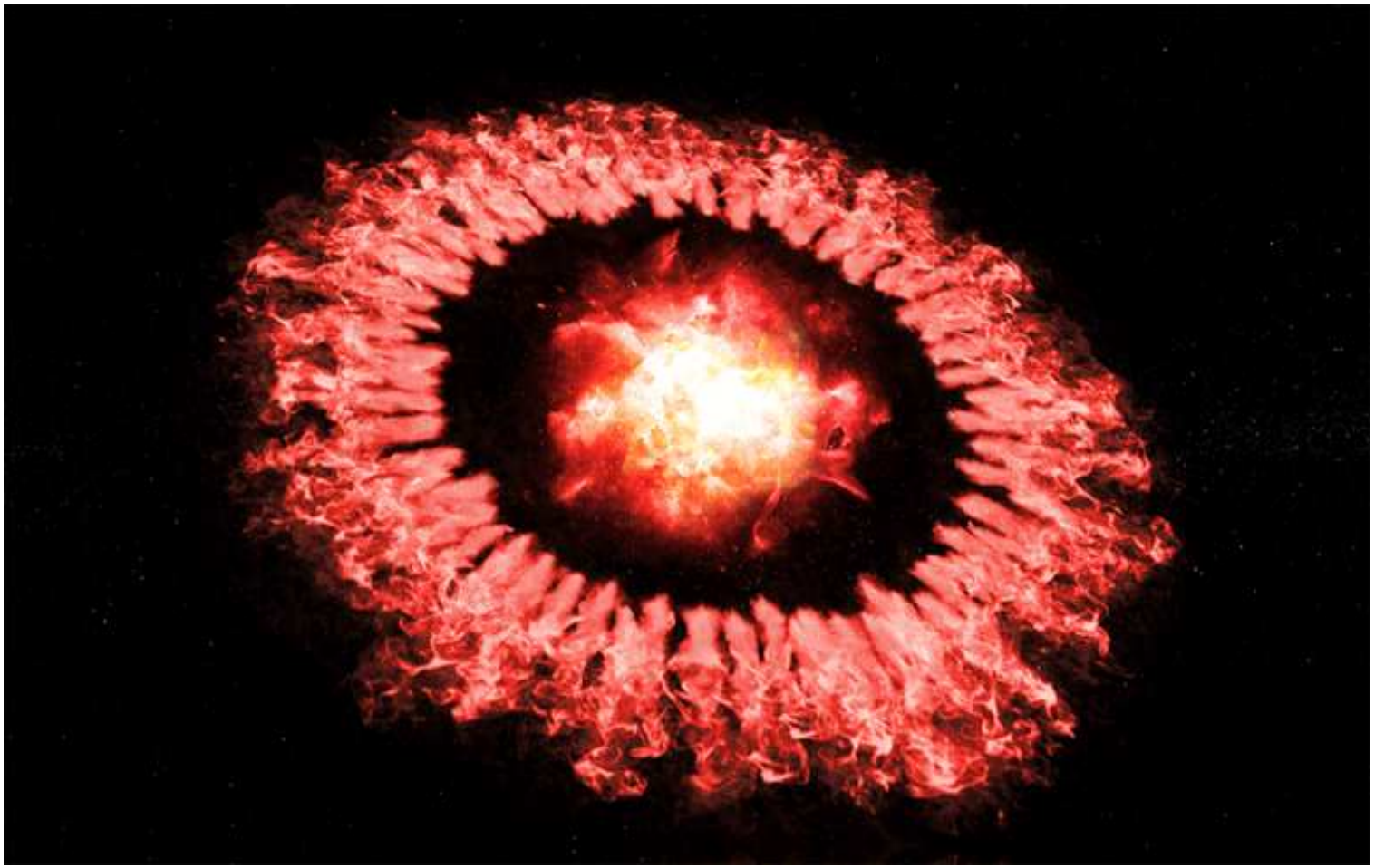
HR Diagram



https://www.secretsofuniverse.in/hertzsprung-russell-diagram/

Death of a star

"To the well-organized mind, death is but the next great adventure" - J. K. Rowling



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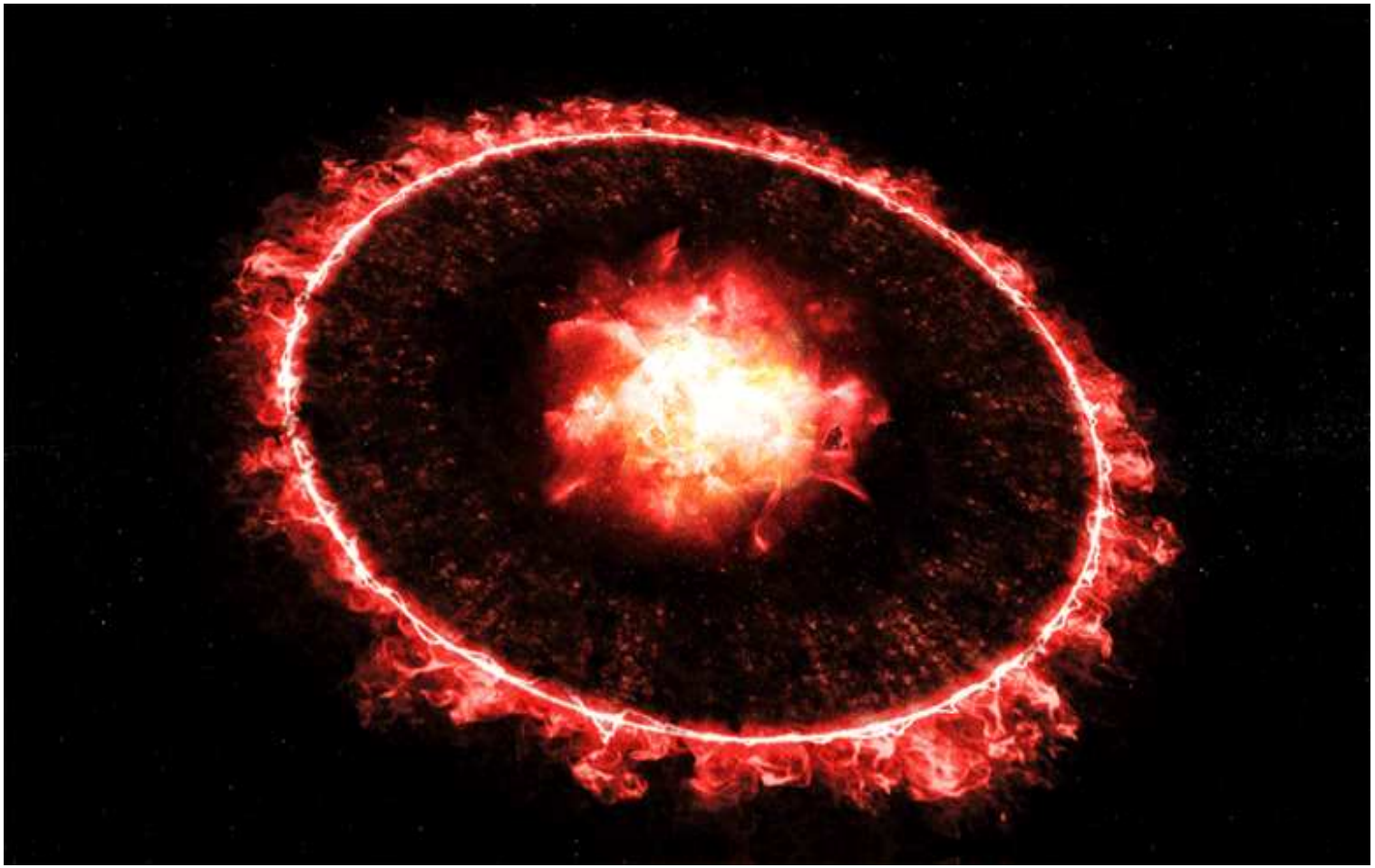

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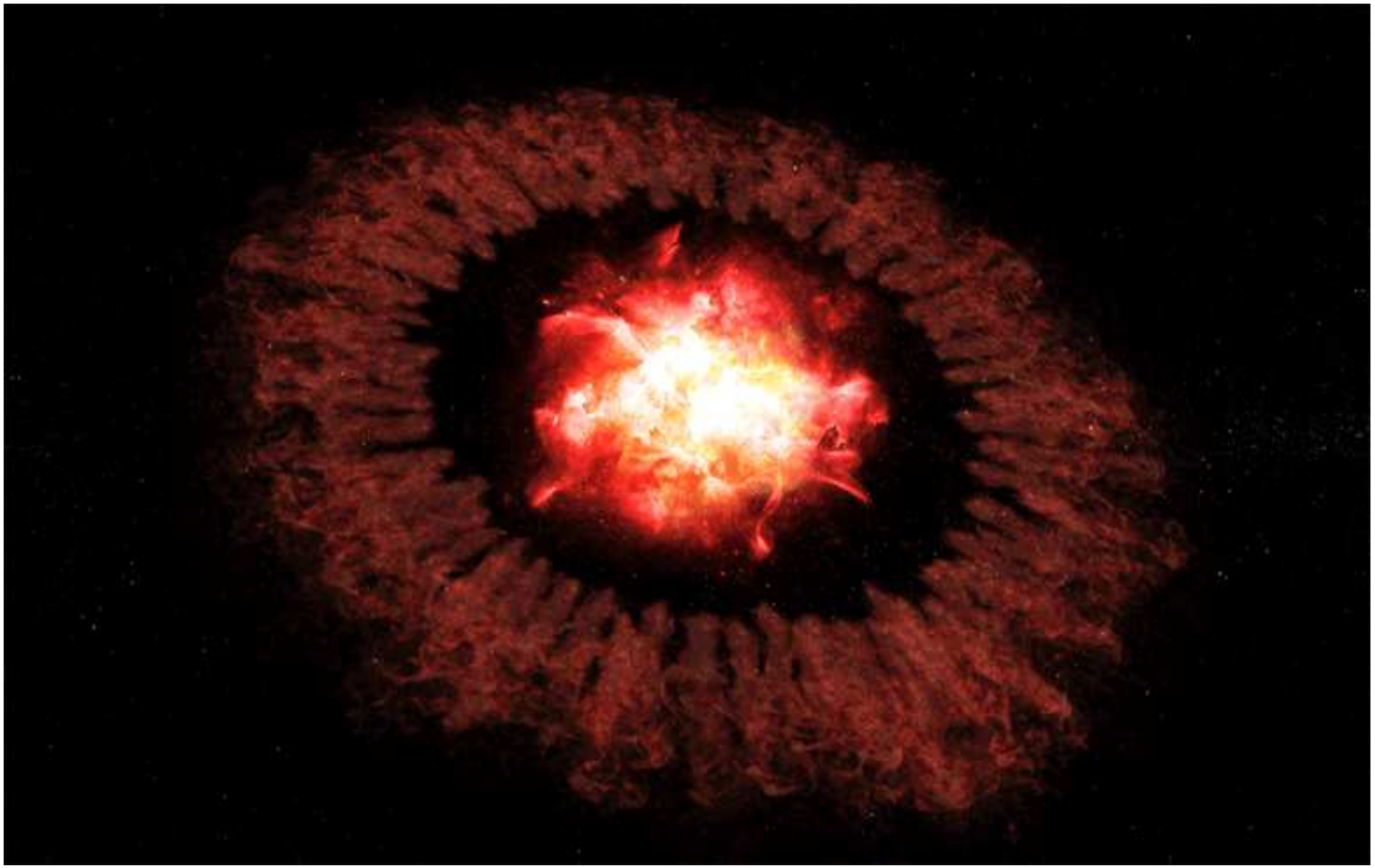

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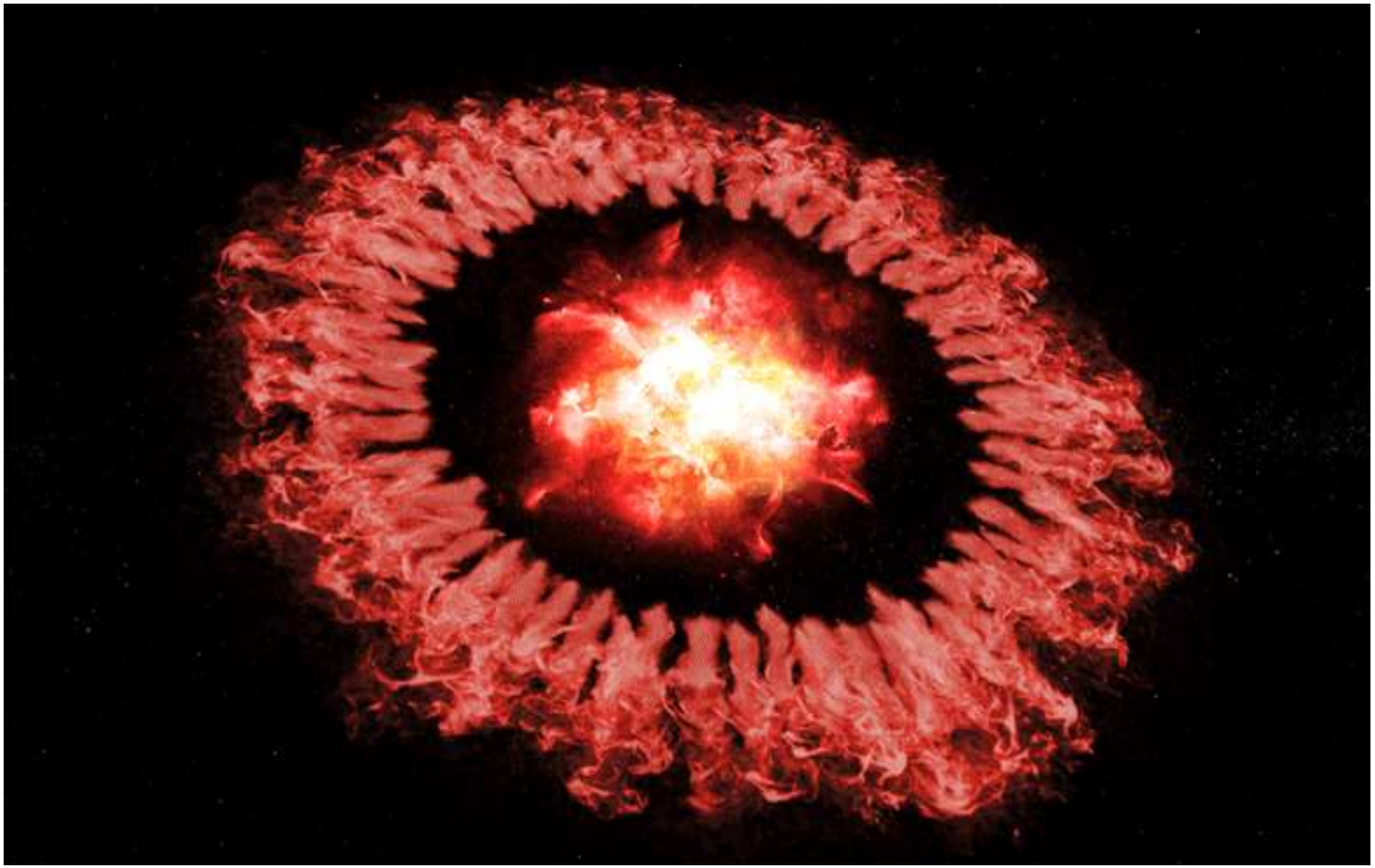

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A neutron star is born

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Pulsar

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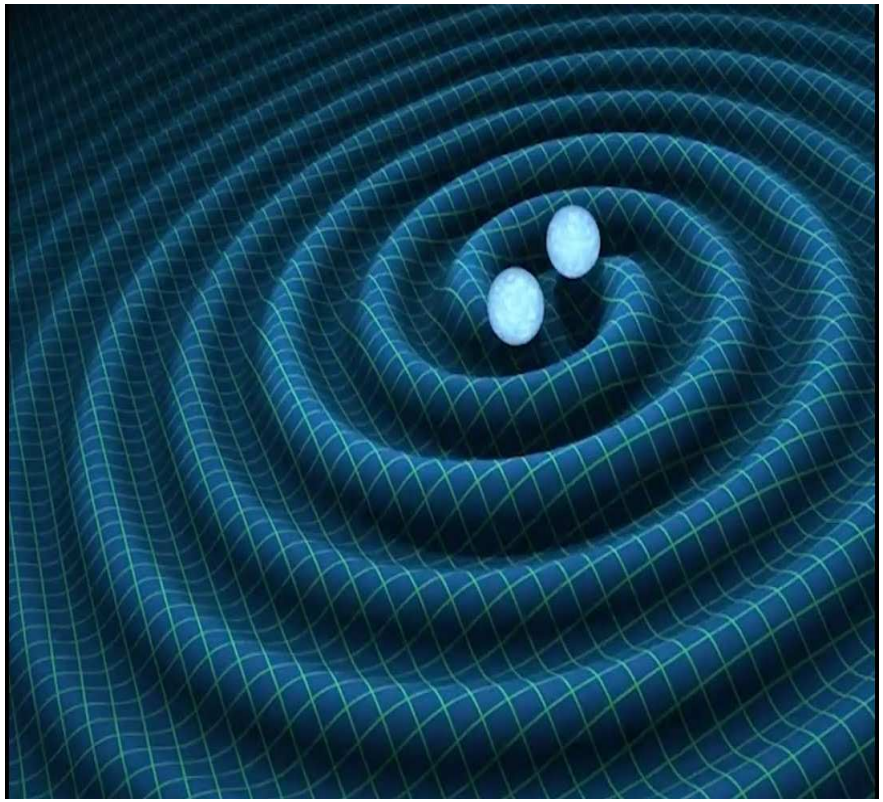

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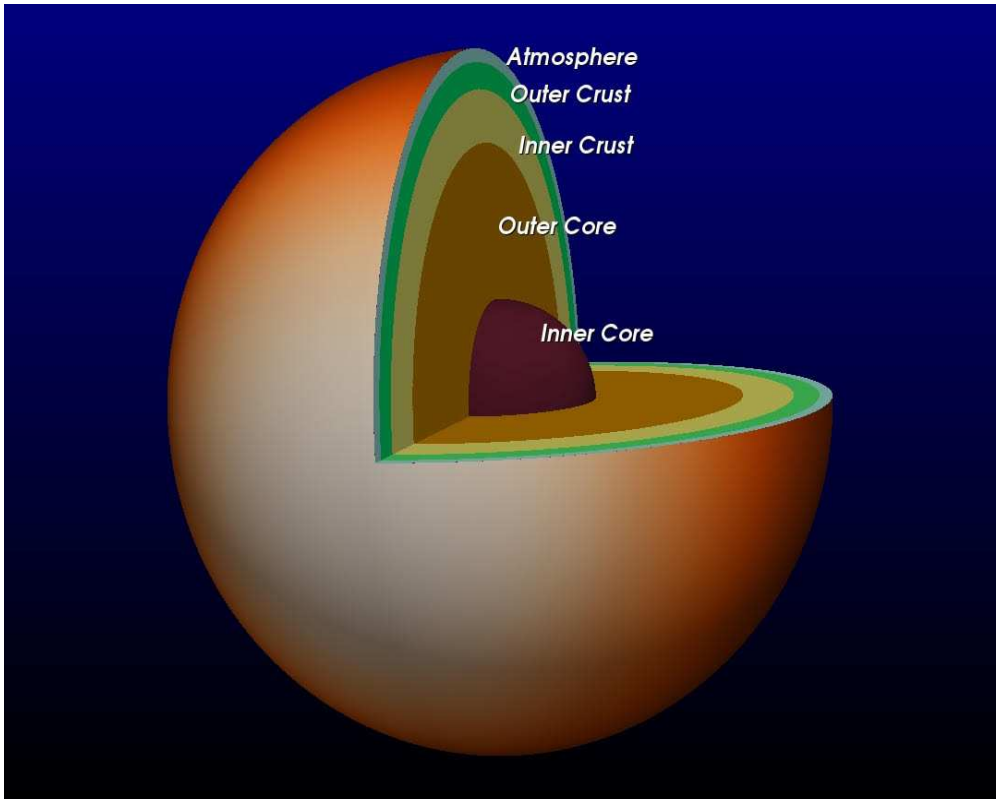
Neutron star



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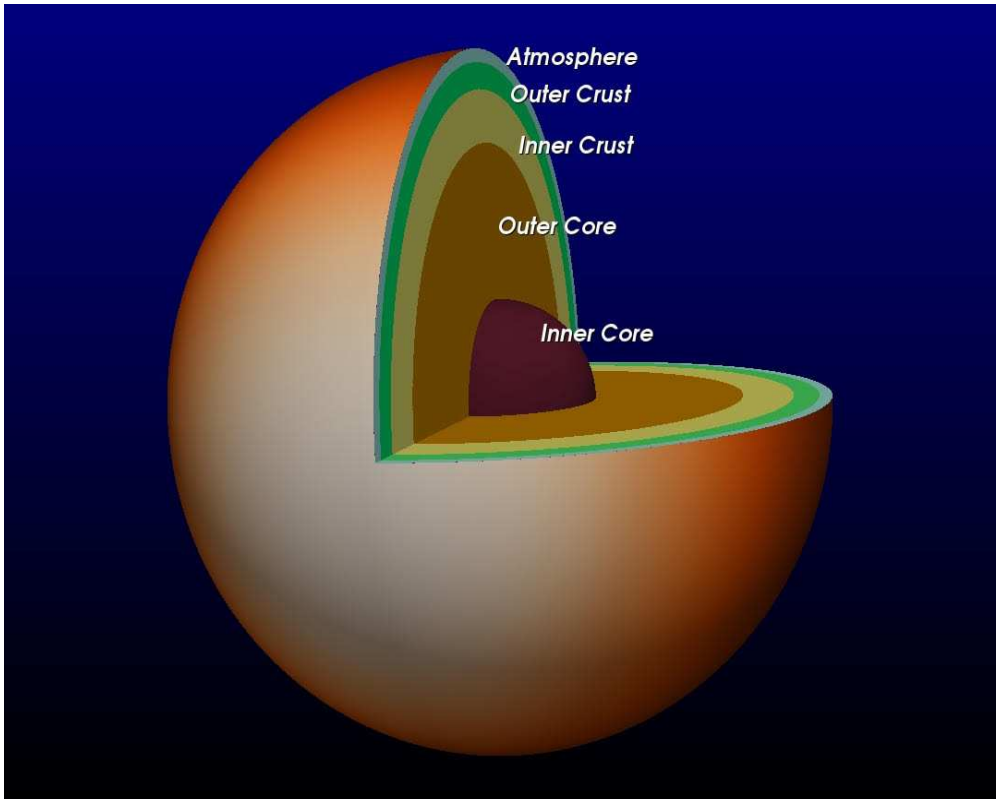

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- A big nucleus?
- Quark Gluon Plasma?
- Phase Transition?
- Radius $\sim$ 10 km
- Density $\sim 10^{15}$ gm.cm^{-3} (5-8 ρ_0)

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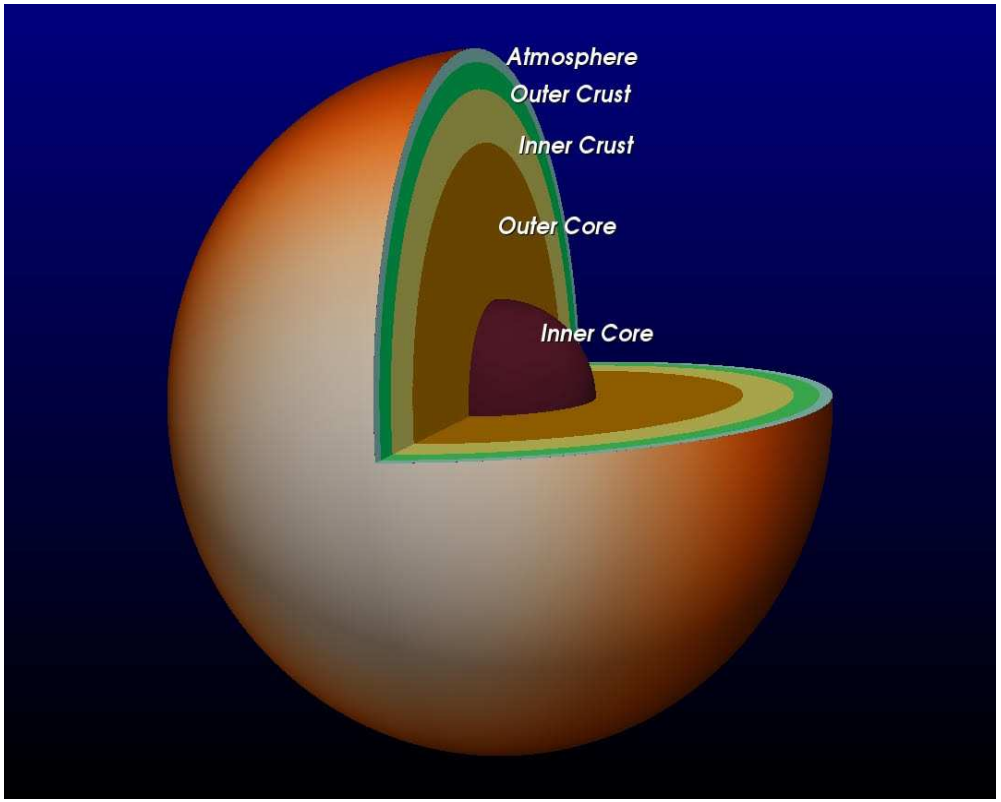

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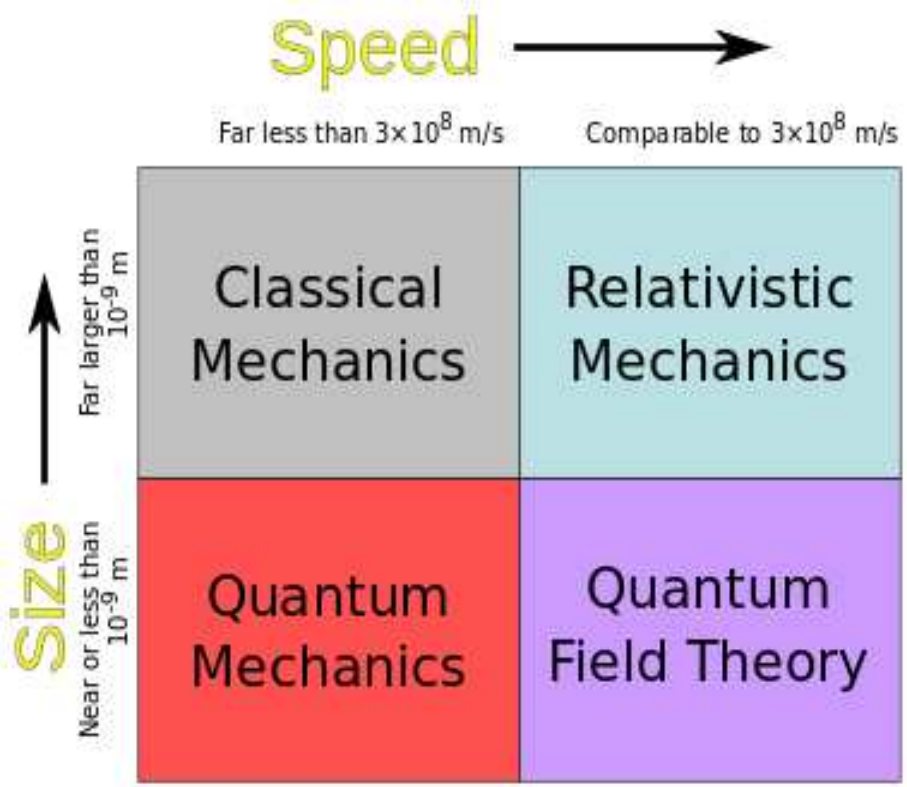
Concerned Forces

- All fundamental forces (Probably)

Different types of dynamics in nature



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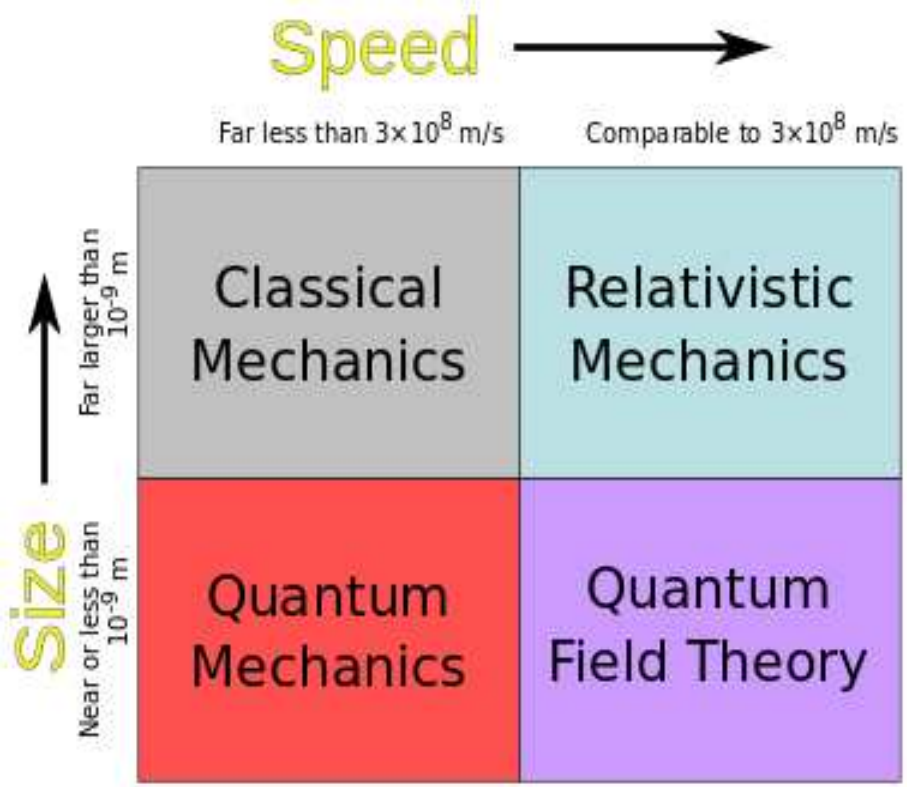
Different types of dynamics in nature



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Field Theory

- Interaction not instantaneous (no potential)
- Naturally incorporates STR
- Can be quantized naturally

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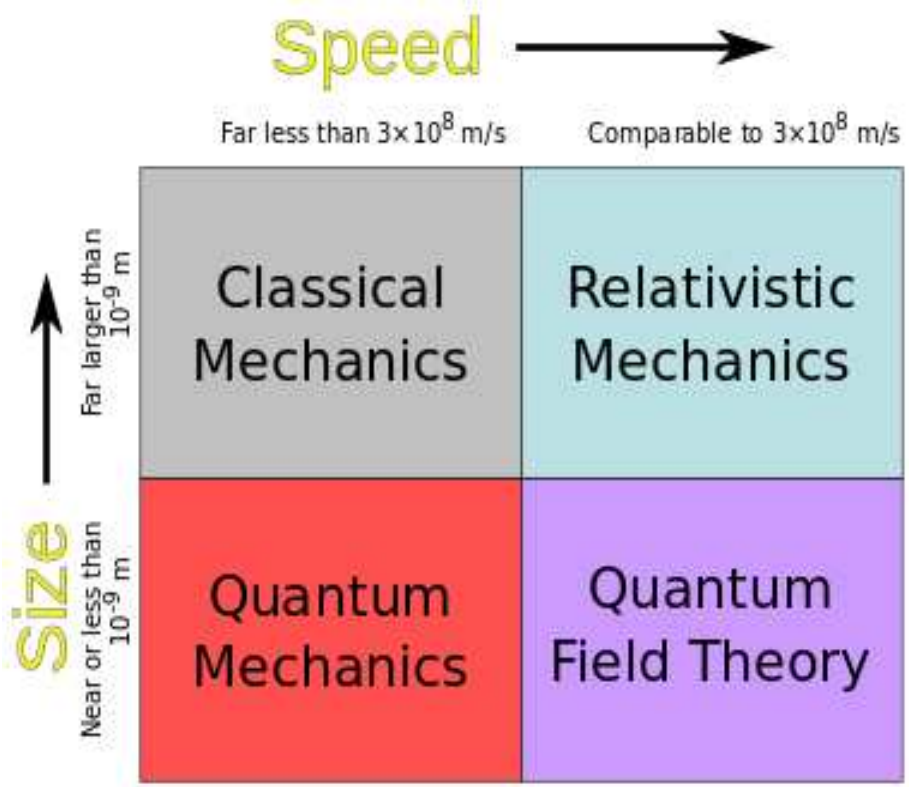

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Field Theory

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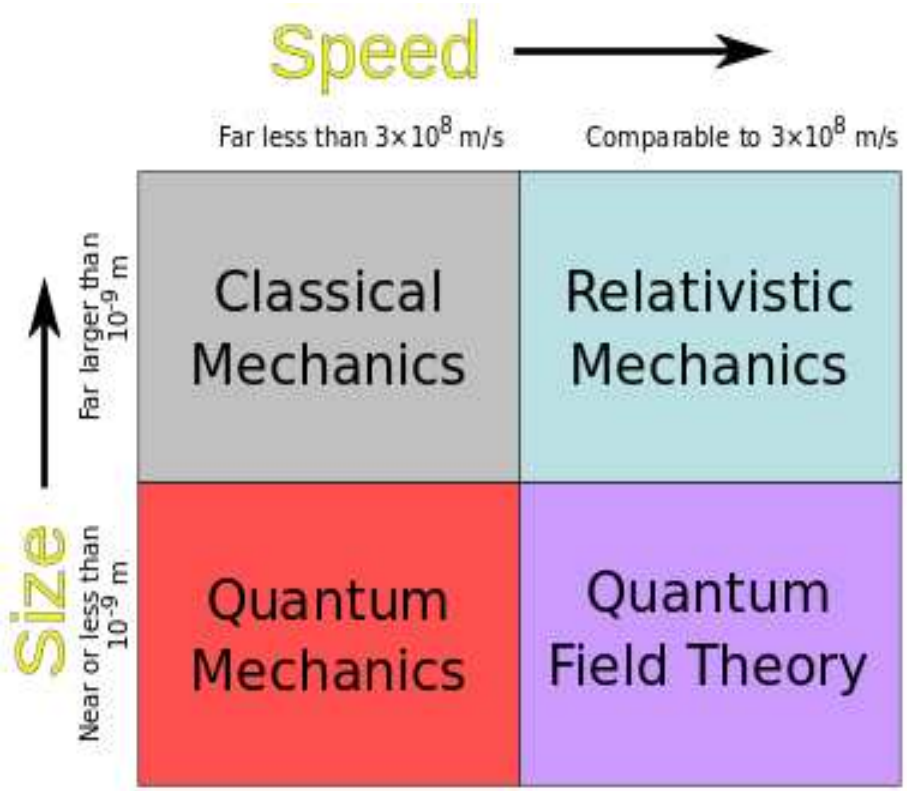
Goal to learn:

- Transformations and Invariances
- Quantization
- Euler-Lagrange Equations
- Energy-Momentum Tensor

Why General Theory of Relativity



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General Theory of Relativity

- Matter and Space are not independent
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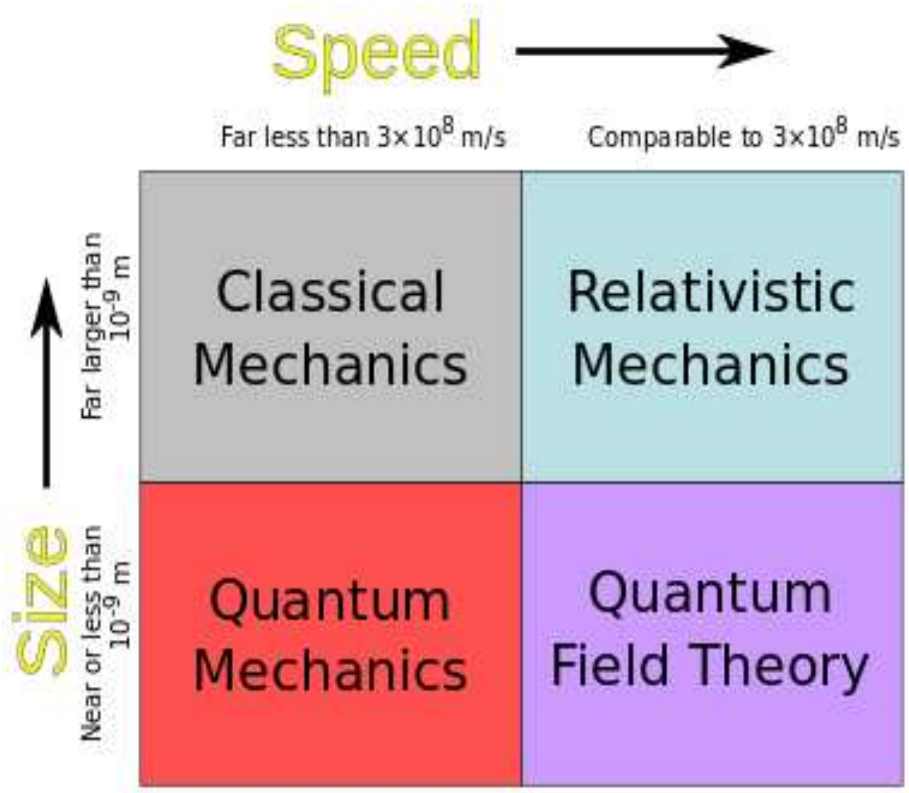

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General Theory of Relativity

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Goal to learn:

- Metrics and Christoffels
- Einstein's Hydrodynamic Equations
- Solving Einstein's Equations in a Spherical system

TOV Equation

$$\frac{dP}{dr} = -\frac{Gm}{r^2}e\left(1+\frac{P}{ec^2}\right)\left(1+\frac{4\pi r^3 P}{mc^2}\right)\left(1-\frac{2Gm}{rc^2}\right)^{-1}$$

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Input:

- Metric
- Equation of state

  $f(e, P) = 0$
- Nuclear Physics Input

  Can experiments in terrestrial condition help?

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Neutron Star

- Mass-Radius Relation
- Maximum mass of a Neutron Star
- New Observations

From nucleus to Neutron Stars

Infinite Nuclear Matter

From nucleus to Neutron Stars

Infinite Nuclear Matter

- Energy per particle of nuclear matter of density $\rho = \rho_n + \rho_p$ and asymmetry $\delta = (\rho_n - \rho_p)/\rho$ is given by,

$$\mathcal{E}(\rho, \delta) = \mathcal{E}(\rho, 0) + \mathrm{E}_{\mathrm{sym}}(\rho)\delta^2 + \mathcal{O}(\delta^4),$$

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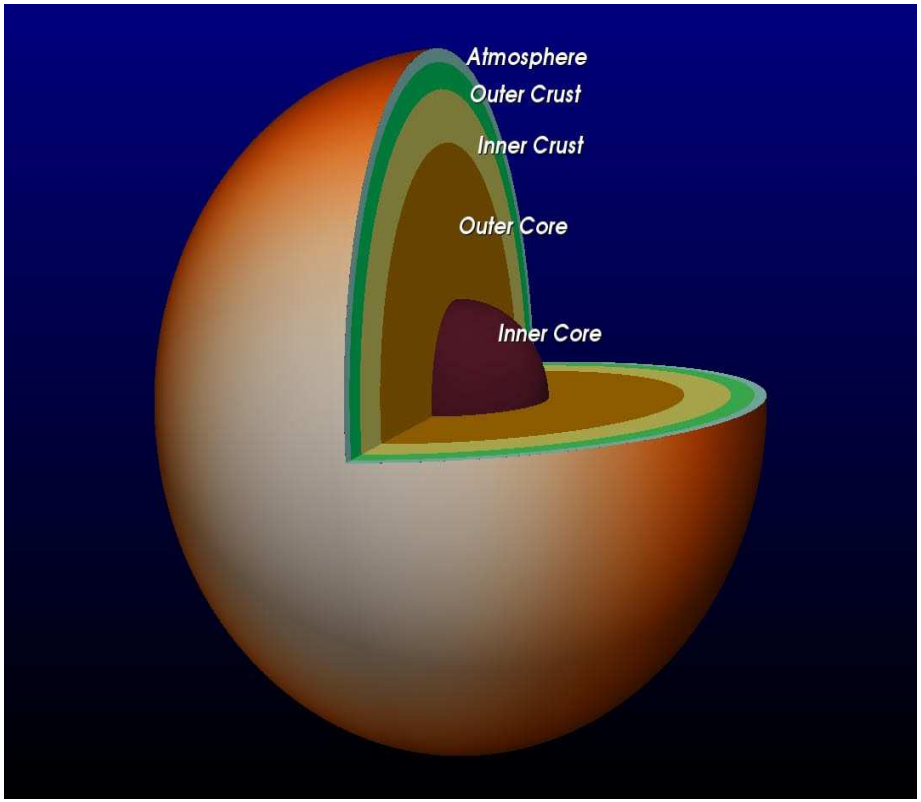
$$\mathrm{P}(\rho, \delta) = \rho^2 \frac{\partial \mathcal{E}(\rho, \delta)}{\partial \rho}$$

Unifying Equation of State

Nuclear Chart helps!!!



Unifying Equation of State

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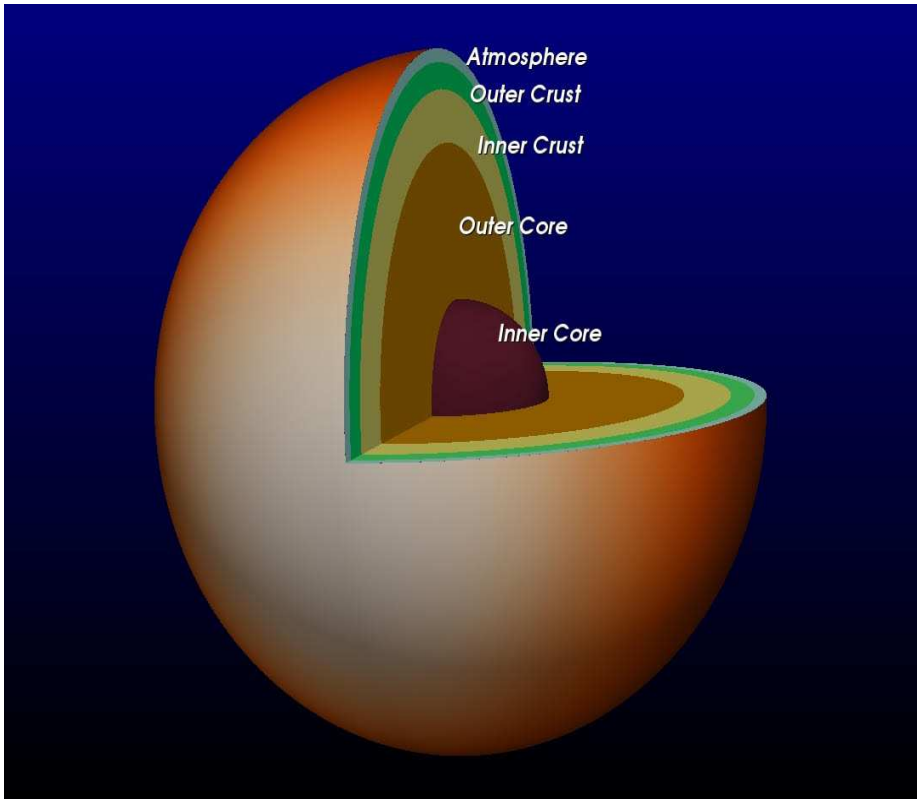

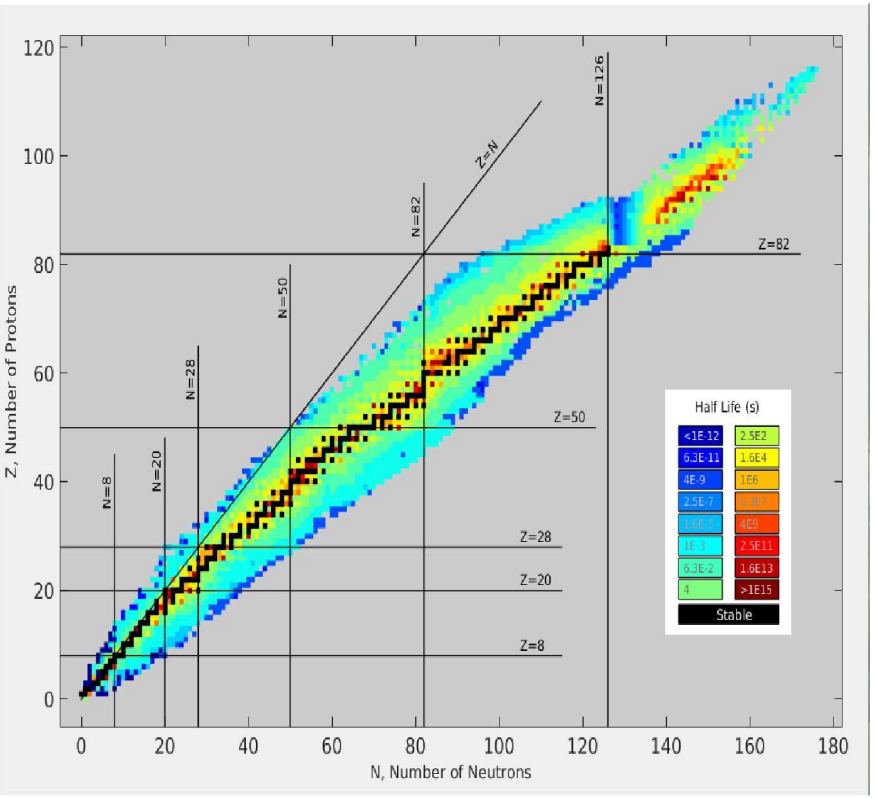

Questions and Comments ...

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